

Short Summary of Numerical Methods

Root Finding Methods

1. Iteration Method

- Formula: $x_{i+1} = g(x_i)$
- Stop when $|x_{i+1} - x_i| < \epsilon$
- Simple but may diverge if $|g'(x)| > 1$.

2. Bisection Method

- Require $f(a)f(b) < 0$
- Formula: $c = \frac{a+b}{2}$
- Replace the endpoint where sign changes.
- Always convergent (slow).

3. False Position (Regula Falsi)

- Require $f(a)f(b) < 0$
- Formula: $c = \frac{af(b)-b(a)}{f(b)-f(a)}$
- Faster than bisection (linear convergence).

4. Newton-Raphson Method

- Formula: $x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$
- Fast (quadratic convergence).
- Needs derivative and good initial guess.

Equation Solving

1. Cramer's Rule

- For system $AX = B$
- $x_i = \frac{\det(A_i)}{\det(A)}$
- Simple for small systems (inefficient for large).

2. Gauss Elimination

- Converts matrix to upper triangular form.
- Then uses **back substitution**.
- Direct method, accurate but computationally heavy.

3. Gauss-Jordan Elimination

- Converts matrix to **reduced row-echelon form**.
- No need for back substitution.
- Gives direct solution.

Value Estimation (Interpolation)

1. Finite Difference (Diagonal & Horizontal)

- Based on equal-interval data.
- **Forward/Backward/Divided** differences used.
- Polynomial interpolation from tabular data.

2. Lagrange Interpolation

- Formula:

$$f(x) = \sum_{i=0}^n f(x_i) \prod_{j=0, j \neq i}^n \frac{(x - x_j)}{(x_i - x_j)}$$

- Works for unequal intervals.

Integration Approximation

1. Trapezoidal Rule

- Formula:

$$I = \frac{h}{2} [f(x_0) + 2\sum f(x_i) + f(x_n)]$$

- Approximates area using trapeziums.

2. Simpson's 1/3 Rule

- Formula:

$$I = \frac{h}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + f(x_n)]$$

- Requires even number of intervals.
- More accurate than trapezoidal.

3. Simpson's 3/8 Rule

- Formula:

$$I = \frac{3h}{8} [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)]$$

- For 3 equal subintervals (n multiple of 3).

Ordinary Differential Equations (ODE)

1. Euler's Method

- Formula: $y_{i+1} = y_i + hf(x_i, y_i)$
- Simple, less accurate (first order).

2. Picard's Method

- Successive approximation:

$$y_{n+1} = y_0 + \int f(x, y_n) dx$$

- Iterative improvement of solution.

3. Runge-Kutta Method (RK4)

- Formula:

$$y_{i+1} = y_i + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

- Very accurate (fourth order).
- Most commonly used.

4. Milne's Method

- Predictor-Corrector approach.
- Uses previous four points.
- Multistep method.

Regression

1. Least Squares Regression

- Minimizes $\sum (y_i - \hat{y}_i)^2$
- Gives best-fit line or curve.

2. Linear Regression

- Model: $y = a + bx$
- Finds line that best fits data.
- $b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$

3. Polynomial Regression

- Fits higher-degree polynomial:

$$y = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

4. Stochastic Gradient Descent (SGD)

- Iterative method to minimize loss:

$$\theta = \theta - \eta \nabla L(\theta)$$

- Used in ML for optimization.

5. Logistic Regression

- Used for classification.
- Formula:

$$P(y = 1 | x) = \frac{1}{1 + e^{-(b_0 + b_1x)}}$$

- Output between 0 and 1.
- Uses **sigmoid function**.
- Loss function: **Log loss (cross-entropy)**.